

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10: SIMILARITY AND ENLARGEMENT (FINAL EXPLANATION)

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation of the sufuria puzzle from Mama Achieng's stall at Kibuye Market in Kisumu. Use what you measured, drew and calculated in Lessons 1 to 8. Write your answers in the space under each prompt. Use $\pi = 3.14$. Show your working for every calculation and include units (cm, cm², cm³, KSh). Where a question asks you to explain, give a reason in full sentences and use the words similar, scale factor k , area and volume. Round only at the end and say 'about'.

Part 1: Are the three sufurias similar?

Mama Achieng sells githeri in a small (S), medium (M) and large (L) sufuria. S has height 12 cm and diameter 16 cm, M has height 18 cm and diameter 24 cm, and L has height 24 cm and diameter 32 cm.

(a) Calculate the ratio of the heights and the ratio of the diameters of M to S and of L to S.

(b) Use your results to explain whether the three sufurias are similar. In your answer, say what must be true of corresponding angles and corresponding sides.

(c) In Lesson 2, rectangle A was 4 cm by 6 cm and rectangle C was 6 cm by 8 cm. Explain why C is not similar to A even though both are rectangles.

(d) What is the linear scale factor k from S to L?

(a) M to S: height $18/12 = 1.5$ and diameter $24/16 = 1.5$. L to S: height $24/12 = 2$ and diameter $32/16 = 2$. So the heights and diameters are both in the ratio 1 : 1.5 : 2.

(b) Similar figures have equal corresponding angles and all corresponding sides in the same ratio (a constant ratio). Every length of M is 1.5 times the length of S and every length of L is 2 times the length of S, so the three sufurias are similar.

(c) Rectangle C to A gives $6/4 = 1.5$ for one pair of sides but $8/6 = \text{about } 1.33$ for the other pair. The ratios are not the same, so C is not similar to A. All angles are right angles, but equal angles alone are not enough; the sides must be in a constant ratio.

(d) $k = 24/12 = 32/16 = 2$.

Part 2: Enlargement and the scale factor

Triangle ABC has AB = 3 cm, BC = 4 cm and AC = 5 cm. It is enlarged from a centre O to give image A'B'C' with A'B' = 6 cm, B'C' = 8 cm and A'C' = 10 cm.

- (a) How can you find the centre O if you only have the drawing of the object and the image?
- (b) Calculate the scale factor k from the sides.
- (c) If OA = 4 cm, what is OA'? Show how you know.
- (d) Triangle ABC is now enlarged from O with k = 0.5. Give the three side lengths of the image and say whether the image is bigger or smaller than the object and whether it is nearer to or further from O.
- (e) How is the triangle enlargement linked to the sufuria S and L?

- (a) Join A to A', B to B' and C to C' with straight lines. The lines meet at one point, and that point is the centre O.
- (b) $k = \text{image length} / \text{object length} = 6/3 = 8/4 = 10/5 = 2$.
- (c) $OA' = k \times OA = 2 \times 4 = 8$ cm.
- (d) The sides are $3 \times 0.5 = 1.5$ cm, $4 \times 0.5 = 2$ cm and $5 \times 0.5 = 2.5$ cm. The image is smaller than the object and nearer to O. The shape is still similar.
- (e) Going from S to L, every length (height 12 to 24 cm, diameter 16 to 32 cm) is multiplied by 2, so L is an enlargement of S with $k = 2$, just like the triangle.}

Part 3: Why paint costs rise faster than the price

The outside area of a sufuria to be painted is the side plus the base, $A = 2\pi r h + \pi r^2$.

- (a) Calculate the outside area of S (radius 8 cm, height 12 cm) and of L (radius 16 cm, height 24 cm).
- (b) Divide the area of L by the area of S. How does this number relate to $k = 2$?
- (c) In Lesson 5 you enlarged a 3 cm by 2 cm rectangle with $k = 3$. Use it to explain why the area changes by this factor and not by k.
- (d) Paint costs about KSh 1 for

- (a) S: $2 \times 3.14 \times 8 \times 12 = 602.88$ cm² and $3.14 \times 8 \times 8 = 200.96$ cm², so $A = 803.84$ cm². L: $2 \times 3.14 \times 16 \times 24 = 2411.52$ cm² and $3.14 \times 16 \times 16 = 803.84$ cm², so $A = 3215.36$ cm².
- (b) $3215.36 / 803.84 = 4$, which is $2^2 = k^2$.
- (c) The rectangle goes from 3 cm by 2 cm (area 6 cm²) to 9 cm by 6 cm (area 54 cm²). $54/6 = 9 = 3^2$. Area depends on two lengths, and both are multiplied by k, so the area is multiplied by $k \times k = k^2$.
- (d) S: $803.84/40$ is about KSh 20. L: $3215.36/40$ is about KSh 80. L costs 4 times as much to paint, which is more than double.}

every 40 cm ² . Find the approximate paint cost for S and for L and say how many times more L costs.	
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Part 4: Why the large sufuria holds so much more githeri

(a) Calculate the capacity of S and of L using $V = \pi r^2 h$. Divide the capacity of L by that of S. (b) In Lesson 6 you found that cubes of side 5 cm and 10 cm have volumes 125 cm³ and 1000 cm³. Use this to explain why the volume scale factor is k^3. (c) One serving of githeri is 400 cm³. Find about how many servings S and L hold when filled to the brim. (d) The price of S is KSh 60 and the price of L is KSh 120. Find the price per serving for each. Who gets the better deal, and why?	(a) S: $3.14 \times 64 \times 12 = 2411.52$ cm³ (about 2.4 litres). L: $3.14 \times 256 \times 24 = 19292.16$ cm³ (about 19.3 litres). $19292.16 / 2411.52 = 8$, so L holds exactly 8 times S. (b) The cube grows from 125 cm³ to 1000 cm³, and $1000/125 = 8 = 2^3$. Volume depends on three lengths, and each is multiplied by k, so the volume is multiplied by $k \times k \times k = k^3$. (c) S: $2411.52 / 400$ is about 6 servings. L: $19292.16 / 400$ is about 48 servings. 48 is 8 times 6. (d) S: $60/6 =$ KSh 10 per serving. L: $120/48 =$ KSh 2.50 per serving. Customers who buy L get the better deal because the price follows only $k = 2$ ($60 \times 2 = 120$) while the capacity follows $k^3 = 8$.)
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Part 5: Final Explanation and prediction

Write your Final Explanation that answers the driving question: Why does a sufuria that is twice as tall and twice as wide hold far more than twice the githeri, and how can we predict how length, area and volume change when we enlarge an object? In your answer: (a) State the scale factor and say what happens to length, area and volume. (b) Explain why each quantity	(a)-(c) The large sufuria is similar to the small one with $k = 2$, because every length is doubled. Lengths multiply by k, areas by k^2 and volumes by k^3. A length depends on one length, an area on two lengths and a volume on three lengths, and each of these lengths is multiplied by k. The capacity of the sufuria depends on three lengths (radius, radius, height), so it multiplies by $2^3 = 8$: L holds exactly 8 times S, about 48 servings against 6. The paint area depends on two lengths, so it multiplies by $2^2 = 4$, and paint costs about KSh 80 against KSh 20. The price only follows $k = 2$ (KSh $60 \times 2 =$ KSh 120). So customers who buy L get a bargain (KSh 2.50 per serving against KSh 10), while Mama Achieng's paint costs rise faster than her price. (d) Volume factor $= 1.5^3 = 3.375$, so the volume is $502.4 \times 3.375 = 1695.6$ cm³. Check: the new radius is 6 cm and height 15 cm, so $V = 3.14 \times 36 \times 15 = 1695.6$ cm³. The two answers agree. (e) The area factor is $3^2 = 9$. The photo area is $10 \times 15 = 150$ cm², so the poster area is $150 \times 9 = 1350$ cm² (30 cm by 45 cm). If the area is 9 times bigger, $k^2 = 9$, so $k = 3$.)
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changes as it does by saying how many lengths it depends on. (c) Link your explanation to the capacity, the paint cost and the price of Mama Achieng's sufurias. (d) A cylinder tin has radius 4 cm and height 10 cm (volume 502.4 cm³). A larger similar tin has $k = 1.5$. Predict the volume of the larger tin using a scale factor, then check it with $V = \pi r^2 h$. (e) A photo 10 cm by 15 cm is enlarged to a poster with $k = 3$. By what factor does the area change, and what is the poster area? A different enlargement makes the area 9 times bigger; what is k?	
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Testing and describing similarity	Calculates all ratios correctly, states that corresponding angles are equal and all sides are in a constant ratio, and explains clearly why rectangle C is not similar to A.	Calculates most ratios correctly and states the constant-ratio idea, with a small gap in explaining why C is not similar.	Relies on shape looking the same or on equal angles only, and does not use a constant ratio of sides.
Scale factor and enlargement	Finds k from lengths and from OA'/OA , finds the centre O by joining corresponding points, and handles both $k = 2$ and $k = 0.5$ correctly.	Finds k correctly and applies it to lengths, with minor errors in describing the centre or the reduction.	Confuses k with the difference between lengths or cannot find the image lengths from k .
Calculating area and volume	Uses $A = 2 \pi r h + \pi r^2$ and $V = \pi r^2 h$ with $\pi = 3.14$ accurately, gets all ratios (4 and 8), and finds servings and costs correctly with units.	Uses the formulas correctly with one or two arithmetic or unit slips.	Uses the wrong formulas or cannot connect the calculated figures to the ratios.
Explaining k , k^2 and k^3	Explains that quantities depend on one, two or three lengths, so they multiply by k , k^2 or k^3 , and uses the cube, rectangle and cylinder evidence. Works	States k^2 and k^3 correctly with some evidence but a limited explanation of why.	Says area and volume simply double when lengths double.

	backwards from an area factor to k.		
Final Explanation of the phenomenon	Links capacity (8 times, about 48 servings against 6), paint (4 times, about KSh 80 against KSh 20) and price (KSh 120 = 2 x 60) to the scale factor, and explains the bargain per serving.	Explains capacity and paint with the correct factors but only partly explains the price or the bargain.	Gives a general answer without the factors or without using Mama Achieng's data.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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